

Preserving Heritage: Enhancing Tourism with AI — Capstone Writeup

Situation

Heritage tourism faces two practical gaps: visitors struggle to identify architectural features in historic sites, and destination platforms need better ways to personalize place recommendations. Indonesia's tourism data (users, places, ratings) plus a multi-class historical-structure image set make this a strong end-to-end AI opportunity—computer vision for structure recognition and collaborative filtering for trip planning.

Task

Deliver a two-part Capstone solution:

1. **Structure classification** — Train a transfer-learning CNN (MobileNetV2) to recognize Gothic/heritage architectural elements (altar, apse, bell tower, column, domes, flying buttress, gargoyle, stained glass, vault), compare training with and without augmentation, and save the best model.
2. **Tourism intelligence** — Clean and explore Indonesian tourism datasets, answer business questions (origins, categories, nature-friendly cities, most-loved spots), and build a collaborative filtering recommender that suggests related places from a seed attraction.

Action

****Part 1 — Vision****

- Organized train/test directories and documented class counts; used a ****stratified subset (~350 images/class)**** for practical CPU training while evaluating on the provided test set.
- Plotted 9 sample images per class with OpenCV + Matplotlib (``outputs/samples_*.png``).
- Built MobileNetV2 (ImageNet weights), ****froze convolutional base****, added Dense(256) + Dropout + Softmax head.
- Compiled with Adam, sparse categorical cross-entropy, and accuracy; added a callback to stop when validation accuracy reaches ****0.85****.
- Trained without augmentation, then with RandomFlip/Rotation/Zoom/Contrast; plotted train/val curves; saved the best checkpoint under ``models/``.

****Part 2 — Tourism + recommender****

- Loaded ``user.csv``, ``tourism_rating.csv``, and ``tourism_with_id.xlsx``; removed duplicate keys and unnamed columns.
- Analyzed age distribution and tourist origin locations; mapped spot categories and city coverage.
- Identified the strongest city for nature enthusiasts using ****Cagar Alam**** volume × rating.
- Merged ratings with place metadata to rank loved places, cities, and categories.

- Built an item–item collaborative filter with `sklearn.NearestNeighbors` (cosine) on the user–item matrix; generated recommendations from seed places such as *Monumen Nasional*.

Result

- Delivered an executed notebook (`preserving\_heritage.ipynb`), metrics JSON, accuracy curves, sample grids, and a saved Keras classifier in `models/`.

- **Final best validation accuracy: 92.53%** (no-augmentation MobileNetV2; exceeded 0.85 target by epoch 1 and continued to epoch 3 for stable curves).

- With-augmentation run: **89.6% → 91.4%** across 3 epochs (also above 0.85). On this strong ImageNet transfer baseline the non-augmented checkpoint edged out slightly; augmentation still showed a clear upward generalization trend.

- Tourism EDA: top origins include Bekasi / Semarang / Yogyakarta; **best city for nature enthusiasts = Bandung** (most `Cagar Alam` spots × rating); **most liked category = Taman Hiburan**.

- Sample CF recommendations for *Monumen Nasional*: Wisata Mangrove Tapak, Danau Rawa Pening, Museum Sonobudoyo Unit I, Dunia Fantasi, Situ Patenggang.

Key artifacts

Artifact	Path
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Notebook	`preserving_heritage.ipynb`
Best model	`models/heritage_structure_classifier_best.keras`
Metrics	`outputs/part1_metrics.json`, `outputs/part2_summary.json`
Plots	`outputs/curves*.png`, `outputs/samples*.png`, `outputs/eda*.png`
Sample recs	`outputs/sample_recommendations.csv`

Screenshots for LMS

1. Per-class sample image grids
2. Class count bar chart
3. Train/val accuracy curves (no-aug vs aug)
4. Age/origins and category EDA plots
5. Sample recommendation table output

Quantified results (from executed run)

- No-aug best val accuracy: 0.9253

- With-aug best val accuracy: 0.9138

- Final best val accuracy: 0.9253

- Best nature city: Bandung

- Most liked category: Taman Hiburan

- Sample seed place: Monumen Nasional

- Wisata Mangrove Tapak (Semarang, Cagar Alam) sim=0.271
- Danau Rawa Pening (Semarang, Cagar Alam) sim=0.263
- Museum Sonobudoyo Unit I (Yogyakarta, Budaya) sim=0.260
- Dunia Fantasi (Jakarta, Taman Hiburan) sim=0.256
- Situ Patenggang (Bandung, Cagar Alam) sim=0.241